ML: MULTIPLE CHOICE QUESTIONS - 5

1. What is the main purpose of a Recurrent Neural Network (RNN)?

A. Image segmentation
B. Handling sequential data
C. Clustering unlabeled data
D. Feature scaling

2. In an RNN, the hidden state at time step ttt depends on:

A. Only the current input
B. Only the previous hidden state
C. Current input and previous hidden state
D. Future inputs only

3. Which problem commonly affects traditional RNNs during training?

A. Overfitting only
B. Vanishing and exploding gradients
C. Data normalization
D. Under-sampling

4. Which activation function is commonly used in classic RNN hidden layers?

A. ReLU only
B. Sigmoid only
C. Tanh
D. Softmax

5. RNNs are widely used in:

A. Time-series forecasting
B. Speech recognition
C. Language modeling
D. All of the above

6. What does "weight sharing" in RNNs mean?

A. Different weights at each time step
B. Same weights reused across time steps
C. Weights are discarded during training
D. Hidden states are fixed

7. LSTM networks were designed primarily to:

A. Reduce computation cost only
B. Solve vanishing gradient problems
C. Replace CNNs entirely
D. Perform dimensionality reduction

8. Which of the following is NOT a gate in an LSTM cell?

A. Forget gate
B. Input gate
C. Output gate
D. Convolution gate

9. The forget gate in an LSTM determines:

A. Which information to add
B. Which information to remove from cell state
C. Final classification output
D. Learning rate adjustment

10. The cell state in LSTM acts as:

A. Temporary random memory
B. Long-term memory pathway
C. Output classifier
D. Input embedding layer

11. Which activation function is typically used in LSTM gates?

A. Softmax
B. Linear
C. Sigmoid
D. ELU

12. Which architecture generally performs better for long-term dependencies?

A. Traditional RNN
B. Linear Regression
C. LSTM
D. K-Means

13. Transformers mainly rely on:

A. Recurrence
B. Convolution
C. Self-attention mechanism
D. Pooling operations

14. Which paper introduced the Transformer architecture?

A. AlexNet
B. Attention Is All You Need
C. GANs for Vision
D. Deep Residual Learning

15. What is the role of positional encoding in Transformers?

A. Reduce overfitting
B. Encode sequence order information
C. Normalize embeddings
D. Increase batch size

16. In self-attention, each token attends to:

A. Only previous tokens
B. Only future tokens
C. All tokens in the sequence
D. Random tokens only

17. Which components are used in the attention mechanism?

A. Query, Key, Value
B. Mean, Variance, Bias
C. Kernel, Filter, Pool
D. Encoder, Decoder, Optimizer

18. Which model is based on the Transformer architecture?

A. BERT
B. GPT
C. T5
D. All of the above

19. What is a major advantage of Transformers over RNNs?

A. Sequential-only processing
B. Better parallelization during training
C. Smaller datasets required
D. No need for attention

20. A GAN consists of:

A. Encoder and Decoder
B. Generator and Discriminator
C. Actor and Critic
D. Teacher and Student

21. The Generator in a GAN aims to:

A. Detect anomalies
B. Generate realistic fake data
C. Compress images
D. Label training data

22. The Discriminator in a GAN is responsible for:

A. Generating images
B. Removing noise
C. Distinguishing real and fake data
D. Feature normalization

23. GAN training is often described as:

A. Supervised learning
B. Reinforcement learning
C. A minimax game
D. Clustering optimization

24. Which application commonly uses GANs?

A. Image generation
B. Deepfake creation
C. Super-resolution
D. All of the above

25. One major challenge in GAN training is:

A. Feature scaling
B. Mode collapse
C. Data cleaning
D. Underfitting only